

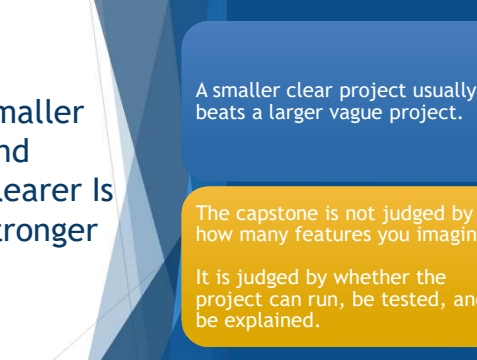


Python Programming Foundations

Week 7 - RBA & Project Framing

Day 2: "Constraints, Structure, & Scope Control"

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Smaller And Clearer Is Stronger

A smaller clear project usually beats a larger vague project.

The capstone is not judged by how many features you imagine.

It is judged by whether the project can run, be tested, and be explained.

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Review: Framing Became Raw Material

Your framing work should now help answer:

- ▶ What is the project for?
- ▶ What does it need?
- ▶ What does it produce?
- ▶ What limits matter?
- ▶ What should AI not decide?

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FROM FRAMING TO PROPOSAL
Yesterday's clarity becomes tomorrow's plan.

A13 FRAMING
Clarify the idea

- PURPOSE**
What is this for?
- INPUTS & OUTPUTS**
What goes in and what comes out?
- CONSTRAINTS & SCOPE**
What are the limits and boundaries?
- LIKELY STRUCTURE**
What kind of solution makes sense?
- RISKS**
What could get in the way?
- AI BOUNDARIES**
What AI should and should not decide?

Your answers become the foundation of your proposal.

A14 PROPOSAL
Shape the plan

- PROJECT GOAL & VALUE**
What are the outcomes and for whom?
- REQUIREMENTS**
Detailed needs and expected outputs.
- SCOPE & CONSTRAINTS**
What is in scope, out of scope, and tested?
- SOLUTION APPROACH**
The structure and key components you will build.
- RISKS & MITIGATIONS**
Key risks and how you will address them.
- AI USE PLAN**
How AI will support the work—and what steps follow.

A focused plan drives results.

Good framing makes proposal writing faster, easier, and stronger. *You already did the hard part!*

Review: Framing Became Raw Material

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What Counts as Success Today

- Identify the realistic core
- Reduce or park extra features
- Choose a structure that fits the project
- Name risks before coding
- Define AI-use boundaries
- Revise the idea into an approvable proposal

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What We Will “Skip” For Now

- ▶ building all features
- ▶ polishing output
- ▶ adding advanced packages
- ▶ packaging for distribution
- ▶ asking AI to keep expanding the project

Today we make the proposal approvable.



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Parked For Later

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Project Scope



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Constraints Are Part Of The Design

"Good constraints make the project stronger by making reality visible."

- time
- skill level
- data access
- explainability
- testing difficulty
- course fit

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CONSTRAINTS SHAPE CLARITY

⇒ Good constraints turn a big idea into a plan you can hit. ⇐

Constraints are not limits. They are design guides that help you succeed.

Constraints Are Part Of The Design

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Feature Count Is Not Project Quality

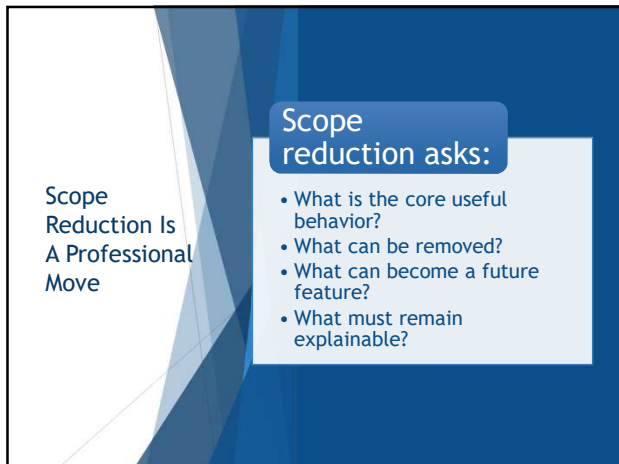
More features can make a project weaker if they make it harder to finish or explain.

Capstone quality comes from a coherent working core.

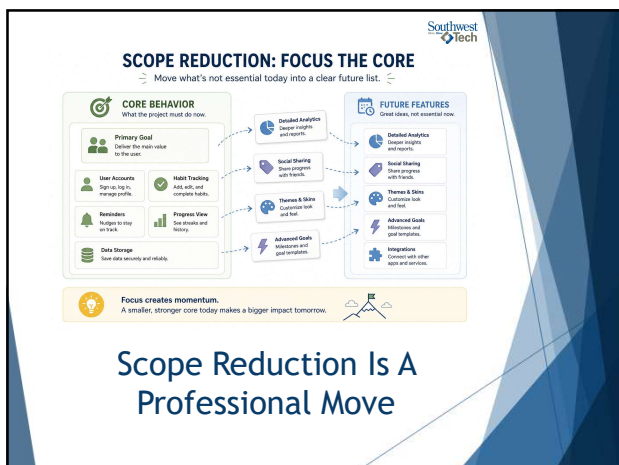
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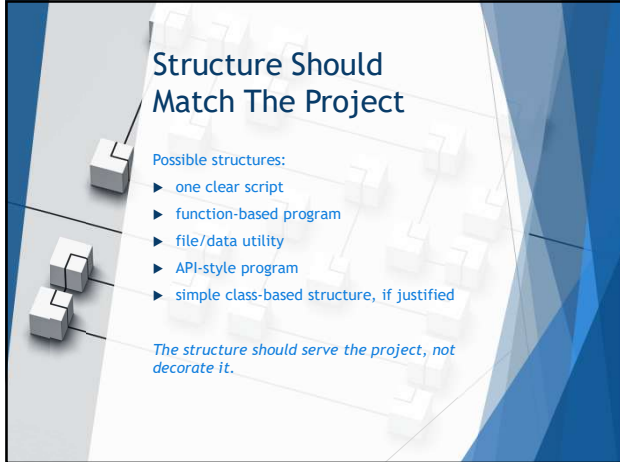
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Structure Should Match The Project

Possible structures:

- ▶ one clear script
- ▶ function-based program
- ▶ file/data utility
- ▶ API-style program
- ▶ simple class-based structure, if justified

The structure should serve the project, not decorate it.

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PATH SELECTION

Different paths. Same goal: a working, explainable project.

Pick the path that fits.

ONE SCRIPT	FUNCTIONS	FILE/DATA UTILITY	API-STYLE PROGRAM	SIMPLE CLASS STRUCTURE
Everything is one file. Good for simple, straightforward tasks.	Break the work into reusable steps. Improves clarity and reduces repeats.	Read, write, and organize data. Useful for local files and datasets.	Expose capabilities through requests. Good when others or other systems will use it.	Group data and behavior together. Helps as complexity grows.

Choose what helps the project finish and explain well.

Structure Should Match The Project Slide

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Risks Should Be Named Before Coding

Risks are things that could make the project harder:

- ▶ live API instability
- ▶ unclear data format
- ▶ too many features
- ▶ logic not yet understood
- ▶ AI code the student cannot explain

Naming risk early makes revision easier.

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RISK PLANNING: PREPARE, DON'T PANIC

Good plans name risks early so you can design around them.

API INSTABILITY

External services may change or fail.

Plan: Use fallbacks and handle errors gracefully.

UNCLEAR DATA

Data definitions or formats may cause confusion.

Plan: Write assumptions and validate early.

PROJECT PLAN

Goal: Build a habit tracker that is useful, clear, and easy to explain.

- 1. USERS
 - Sign up and log in
 - Manage profile
- 2. CORE FEATURES
 - Add only, complete habits
 - View progress
- 3. DATA & STORAGE
 - Track habits and progress
 - Keep data consistent
- 4. INSIGHTS
 - Weekly and history
 - Simple summaries
- 5. POLISH & LAUNCH
 - Basic version
 - Prepare for sharing

TOO MANY FEATURES

Scope creep can dilute the core.

Plan: Focus on the core and park extras for later.

UNEXPLAINED AI CODE

AI suggestions may be hard to understand or maintain.

Plan: Review, simplify, and document clearly.

Risks don't disappear. Planning for them keeps the project solid and the explanation clear.
Name it. Plan for it. Move forward with confidence.

Risks Should Be Named Before Coding

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Demo #1

"Scope Reduction Example"

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What to Watch For

Watch the project shrink and improve.

- ▶ The goal is not to make the idea less interesting.
- ▶ The goal is to make the core project finishable.

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PROJECT SCOPE: FOCUS CREATES IMPACT
From everything on the wishlist to a version that works, wows, and can grow.

BEFORE
BIG PROJECT, MANY FEATURES
 Lots to build. Harder to finish and explain.

AFTER
FOCUSED PROJECT, STRONG CORE
 Clear value. Easier to finish and explain.

ONE STRONG CORE
 Core Purpose
 Help users build habits and save real progress.

LATER LIST
 Great ideas for future versions.

You're not lowering the goal.
 You're choosing the best path to a finish line that matters.

**Demo 1:
 Scope Reduction Example**

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Demo #2
 "Structure Options Comparison"

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What to Watch For

Different structures fit different projects.

The best structure is the one that makes the project easier to:

- Finish
- Test
- ... And explain.

Q2 **Q3**

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STRUCTURE CHOICES: MANY GOOD PATHS

Different structures fit different projects. Choose what helps you build, explain, and maintain your solution.

SCRIPT	FUNCTIONS	DATA UTILITY	API PATH	CLASS-BASED PATH
Everything is one file. Straightforward and easy to run.	Break work into reusable pieces. Improve clarity and reduce repeats.	Focus on reading, writing, and managing data and files.	Expose capabilities through requests. Connect with other systems.	Group data and behavior together. Scale with structure.
Good for: • Simple tasks • Prototypes • One-time tools	Good for: • Repeated logic • Clear organization • Easier testing	Good for: • File and data work • Conversions • Local processing	Good for: • Integrations • Remote access • Service-style apps	Good for: • Complex domains • Many related parts • Long-term growth
Example: data_loader.py	Example: calculator.py	Example: report_builder.py	Example: user_service.py	Example: task_manager.py

There is no single "best" structure.
Pick the path that fits your project, your skills, and your goals.

Demo 2: Structure Options Comparison

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Demo #3

"AI Boundary Example"

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What to Watch For

AI boundaries should be specific.

"I used AI" is too vague.

A useful boundary says

- ▶ what AI may help with
- ▶ what you must decide
- ▶ and what must be verified.

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Proposal Evidence

Your proposal should show:

- ▶ clear project purpose
- ▶ feasible feature list
- ▶ expected inputs and outputs
- ▶ likely structure
- ▶ risks or unknowns
- ▶ AI-use boundaries
- ▶ why the scope is realistic

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PROPOSAL CHECKLIST

A clear plan is easier to explain and easier to approve.

	1. PURPOSE What problem does this project solve and why does it matter?	☐
	2. FEATURES What are the main features? What is in scope and what is out?	☐
	3. INPUTS AND OUTPUTS What inputs will the project use? What outputs will it produce?	☐
	4. STRUCTURE What approach or structure will you use and why does it fit this project?	☐
	5. RISKS What could go wrong? How will you reduce or handle the risks?	☐
	6. AI-USE BOUNDARIES Where will AI help? What decisions will you make and verify?	☐
	7. REALISTIC SCOPE Can this be built, tested, and explained within the time and resources you have?	☐

Keep it focused. Keep it clear. Keep it doable.
A strong proposal turns ideas into action.

Proposal Evidence

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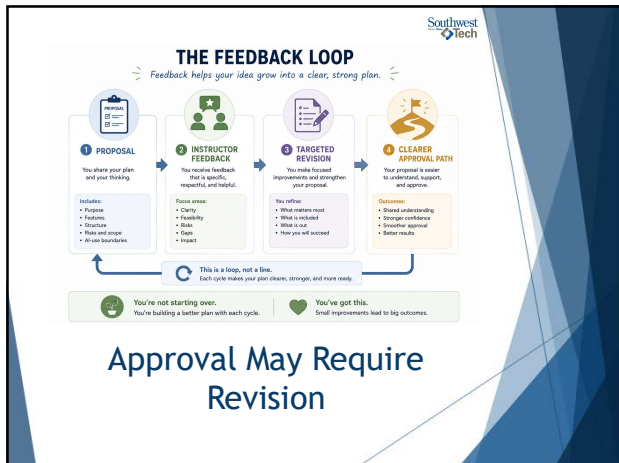
Approval May Require Revision



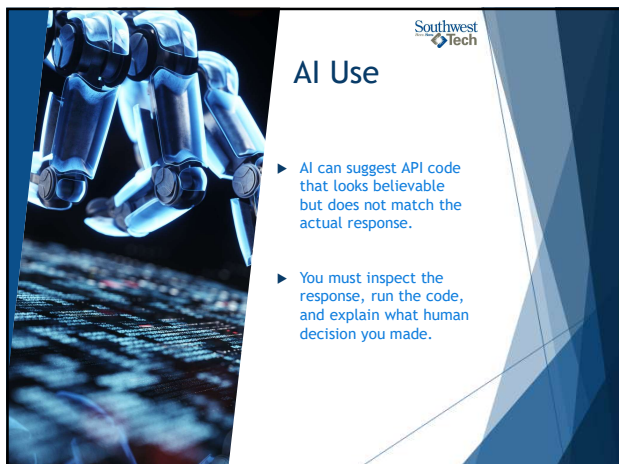
Approval is not just **yes** or **no**.

The instructor may approve, approve with limits, or request targeted revision so the project becomes more realistic.

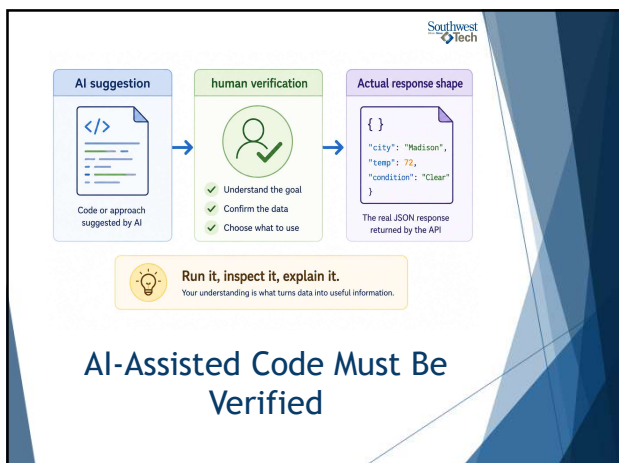
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Closing Success Check

By the end of today, you should be able to say:

"My proposal is smaller, clearer, more realistic, and easier to explain than my first idea."

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From First Idea to Strong Plan

Great projects start big. Strong projects get focused.

FIRST IDEA

Wide and Unclear



Too many ideas. Hard to build. Hard to explain.

REFINE WITH PURPOSE

MAKE IT:

- SMALLER**
Focus on one core problem to solve.
- CLEARER**
Define the purpose, users, and value.
- REALISTIC**
Match the scope to time and resources.
- EXPLAINABLE**
Easy to describe, review, and approve.

STRONGER PLAN

Focused and Ready

Core idea: A simple task tracker for students.

Includes:

- Create tasks
- Mark complete
- View my tasks

Out of Scope (Later):

- Social features
- Payments
- AI suggestions
- Mobile app

Clear focus. Achievable. Easy to explain and approve.

Better plans lead to better builds.
Focus first. Add more later.

Success Check

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Thank you!

Have Fun!

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